

BPM2_NB_ESOFT QUIZ 6_S2026

of Questions: 30

Question #: 1

A 54-year-old man comes to the physician because of a 1-week history of hearing loss in both ears. He has served in the military for the previous 30 years. Physical examination shows sensorineural hearing loss most likely caused by noise-induced trauma to the hair cells of the inner ear. Which of the following best characterizes the localization of these cells in the membranous labyrinth?

- A. Scala vestibuli
- B. Scala media
- C. Scala tympani
- D. Utricle
- E. Sacculle

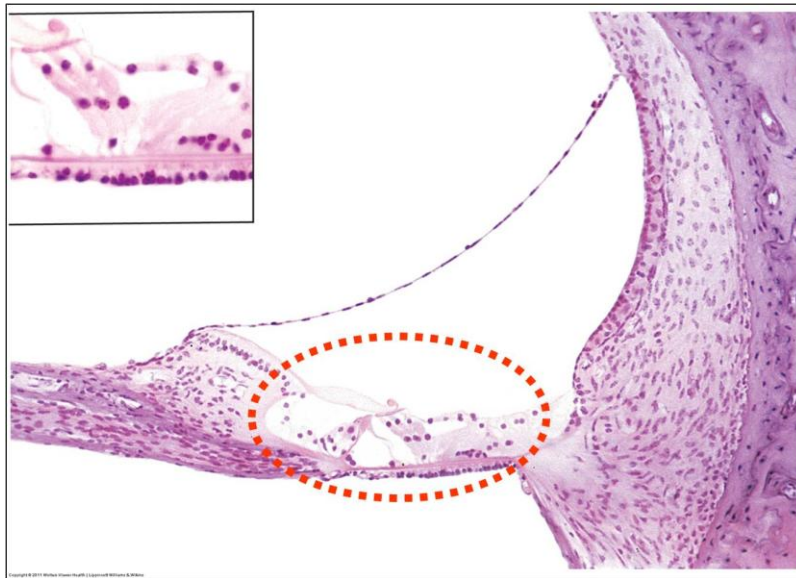
Item Description: HCB.1383 - Membranous labyrinth

Question #: 2

An 8-year-old boy is brought to the physician by his parents because of a 3-month history of worsening hearing loss. Physical examination shows a sensorineural hearing loss of both ears. Urinalysis shows massive proteinuria. The physician suspects an autoimmune destruction to the circumscribed structure shown in the attached photomicrograph. Which of the following structures most likely forms part of the affected tissue?

- A. Spiral ganglion
- B. Vestibular membrane
- C. Tectorial membrane
- D. Stria vascularis
- E. Otolithic membrane

Attachment:



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Item Description: HCB.1383 - Structure of the Organ of Corti

Question #: 3

A 12-year-old boy is brought to the physician by his mother because of a 1-day history of pain in his throat when swallowing. Physical examination shows the posterior aspect of the tongue is swollen and red. Which of the following nerves is most likely responsible for the pain felt by the patient?

- A. Facial
- B. Lingual
- C. Internal laryngeal
- D. Vagus
- E. Glossopharyngeal

Item Description: A 12-year-old boy is

Question #: 4

A 24-year-old woman comes to the physician because of a 3-week history of a continuous painful sensation. She says she has a full feeling in her throat while speaking. Physical examination shows no swelling or redness in the pharynx. Laryngoscopy shows a large growth on the left vestibular fold. The pain from the lesion is most likely transmitted by which of the following nerves?

- A. Recurrent laryngeal
- B. External laryngeal
- C. Internal laryngeal
- D. Glossopharyngeal
- E. Pharyngeal plexus

Item Description: A 24-year-old woman

Question #: 5

A 25-year-old man comes to the physician because of a 2-week history of decreased salivation and flattening of the nasolabial fold. Physical examination shows decreased taste on the anterior third of the tongue and decreased strength in his facial muscles. There are no problems with lacrimation and his nasal cavity is moist. There is no loss of hearing or problems with balance. At which of the following locations is the lesion that resulted in these symptoms most likely located?

- A. Facial nerve in the internal acoustic meatus
- B. Facial nerve in the medial wall of the middle ear
- C. Facial nerve at the stylomastoid foramen
- D. Greater petrosal nerve in the pterygopalatine fossa
- E. Chorda tympani in the infratemporal fossa

Item Description: A 25-year-old man co

Question #: 6

An 18-year-old woman comes to the physician because of a 1-day history of being unable to speak. She attended a sports event 3 days ago during which she became hoarse. Physical examination shows no signs of fever or cough. The physician explains that her condition is due to strain on the muscle responsible for tensing the vocal cords. Which of the following nerves most likely innervates the affected muscle?

- A. Recurrent laryngeal
- B. External laryngeal
- C. Internal laryngeal

- D. Superior laryngeal
- E. Gossopharyngeal

Item Description: An 18-year-old woman

Question #: 7

A 2-month-old boy is brought to the emergency department by his parents because of a sudden onset of seizures. His mother says he has been feeding poorly for the past 2 days. His blood ammonia level is 800 µg/dL (normal range is 15 - 45 µg/dL). His blood urea nitrogen level is low, and urine and serum are positive for orotic acid. Which of the following enzyme deficiency disorders is most likely?

- A. Lysosomal glucosidase
- B. Glucose 6-phosphatase
- C. Medium chain acyl CoA dehydrogenase
- D. UMP synthase
- E. Ornithine transcarbamylase

Item Description: BCHM.MB. 2001: Urea cycle disorder OTC

Question #: 8

A 10-month-old boy is brought to the emergency department by his parents because of a 2-hour history of seizures. Physical examination shows an enlarged liver and kidney. Examination of his heart is within normal limits. Laboratory studies are shown in the table. Urine samples are negative for reducing sugars. Which of the following inherited disorders of metabolism is the most likely diagnosis?

- A. Myopathic carnitine deficiency
- B. Phenylketonuria
- C. Classical galactosemia
- D. Hereditary fructose intolerance
- E. von Gierke disease
- F. McArdle disease
- G. Medium chain acyl CoA dehydrogenase deficiency

Attachment:

Analyte	Patient	Reference range
Glucose	45 mg/dL	70-100 mg/ dL
Lactate	4 mmol/ L	0.5-1 mmol/ L
Arterial pH	7.3	7.35-7.45
Ketone bodies	0.6 mmol/L	0.4-0.5 mmol/L

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Item Description: BCHM.MB.0910; von Gierke disease

Question #: 9

A 14-year-old girl is brought to the physician by her parents because of a 1-week history of increased frequency of urination, increased thirst and weight loss. Physical examination shows dry mucus membranes. Urine dipstick tests are positive for glucose and ketone bodies. Which of the following would most likely be an additional laboratory finding in this patient?

- A. Decreased plasma glucose levels
- B. Normal HbA1c levels
- C. Decreased glucose output by liver
- D. Increased circulating C-peptide levels
- E. Increased anion gap metabolic acidosis
- F. Increased serum bicarbonate levels

Item Description: Type -1 Diabetes mellitus

Question #: 10

A 45-year-old woman comes to the physician because of a 2-month history of reduced hearing in both ears. She was treated with a 2-week regimen of gentamicin (an aminoglycoside antibiotic known to cause vestibular or cochlear ototoxicity) 3 months ago due to a complicated urinary tract infection. Physical examination shows no abnormalities. Audiometry shows sensorineural hearing loss most likely caused by damage to the hair cells of the organ of Corti of the inner ear. Which of the following best characterizes the affected structure?

- A. Lateral surface has both kinocilium and stereocilia
- B. Apical surface contains a single true kinocilium and stereocilia
- C. Apical surface lacks both kinocilium and stereocilia
- D. Kinocilium is absent on the apical surface
- E. Possesses a kinocilium on the basal aspect

Item Description: HCB.1384 - Structure of the Organ of Corti - sensory hair cells

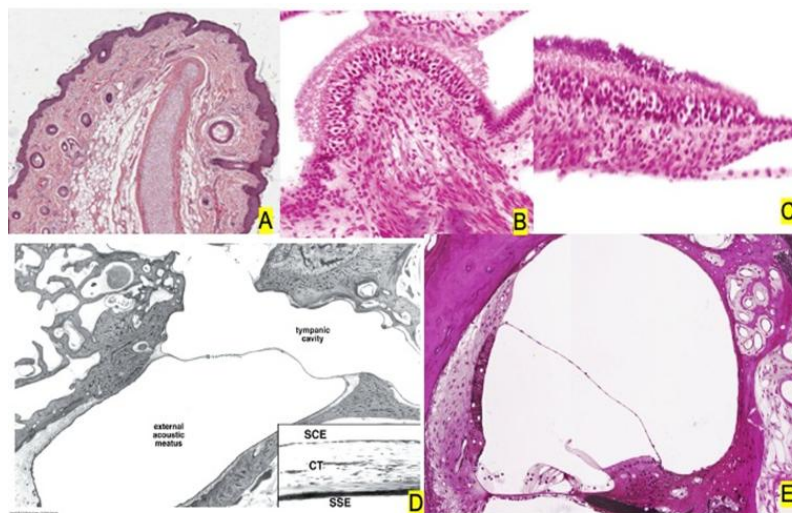
Question #: 11

A 5-year-old girl is brought to the physician by her mother because of a 1-week

history of fever, cough, and discharge from the right ear. Her temperature is 40°C (104°F). Physical examination shows a febrile child with sunken eyes, poor skin turgor, and yellow, viscous discharge from the right ear. Otoscopy of the ear shows a perforation of the tympanic membrane, with a purulent discharge. Which of the following microanatomical photomicrographs most likely shows the perforated structure?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



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Item Description: HCB.1360 - Tympanic membrane

Question #: 12

A 45-year-old man comes to the emergency department because of a 20-min history of bleeding in the scalp following a fall. His vital signs are within normal limits. Physical examination shows a 1-cm lacerated wound in the left temporal region of his scalp. The physician decides to suture this wound. He is administered lidocaine, a local anesthetic that blocks certain voltage-gated ion channels in the sensory nerves supplying the wound area. Which of the following ion fluxes during an action potential in the neuronal membrane is most likely blocked by this drug?

- A. K⁺ influx
- B. K⁺ efflux
- C. Na⁺ efflux
- D. Cl⁻ efflux
- E. Na⁺ influx

Item Description: PHYS.0060 Action Potential block

Question #: 13

A 31-year-old woman comes to the physician because of an 8-month history of irregular menstrual cycles, acne, and darkening of the skin. Her blood pressure is 142/94 mm Hg; other vital signs are within normal limits. Physical examination shows increased hair on the face and hyperpigmentation of skin creases. Laboratory studies show elevated serum insulin, androgens and increased LH/FSH ratio. Pelvic ultrasonography shows cystic enlargement of the ovaries. She is diagnosed with polycystic ovarian syndrome. Which of the following is the most likely source of androgens during the early follicular stage in a normal woman?

- A. Cells of zona glomerulosa of adrenal cortex
- B. Granulosa cells in ovary
- C. Theca cells in ovary
- D. Endometrial lining of the uterus
- E. Corpus luteum of the ovary

Item Description: PHYS.1076. Source of precursors for estrogen synthesis

Question #: 14

A 45-year-old man comes to the physician because of a 2-week history of headache, decreased urination, and muscle cramps. He was diagnosed with small cell carcinoma of the lung 4 months ago. His vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies show elevated serum ADH and atrial natriuretic peptide. He is diagnosed with syndrome of inappropriate ADH (SIADH). Which of the following additional laboratory findings is most likely in this patient?

- A. Low plasma fasting glucose
- B. Low serum osmolality
- C. Elevated serum sodium concentration
- D. Urinary osmolality less than 300 mOsm/kg
- E. Low serum potassium concentration

Item Description: PHYS.1021. SIADH lab findings

Question #: 15

A 34-year-old woman comes to the physician because of a 4-week history of abdominal discomfort and increased bowel movements. She says these symptoms are worse when she postpones her meals. Her vital signs are within normal limits. Physical examination shows no abnormalities. Laboratory studies of a

stool sample show an increased acidity. She is suspected of having deficiency of a hormone that normally helps to neutralize acid in the small intestine. A low level of which of the following hormones is most likely in this patient?

- A. Gastrin
- B. Histamine
- C. Secretin
- D. Somatostatin
- E. Cholecystokinin

Item Description: PHYS.1130. Hormone involved in acid neutralization in small intestine

Question #: 16

A 45-year-old woman comes to the physician because of a 2-month history of difficulty hearing in the right ear. Neurological examination shows: Weber's test lateralizes to the right, Rinne's test BC is greater than AC on the right, AC is greater than BC on the left. Which of the following is the most likely location of the lesion?

- A. Inferior Colliculus
- B. Lateral Lemniscus
- C. Transverse temporal gyrus
- D. Tympanic membrane
- E. Cochlear nerve

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0244 Clinical assessment of auditory function

Question #: 17

A 62-year-old man comes to the physician because of a 2-day history of difficulty controlling the left side of his body. Imaging studies show an infarction affecting the left superior cerebellar peduncle. Which of the following motor systems is most likely dysregulated by the lesion?

- A. Rubrospinal
- B. Lateral vestibulospinal
- C. Medial vestibulospinal
- D. Lateral reticulospinal

E. Medial reticulospinal

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0200 Cerebellar connections to motor centers

Question #: 18

A 27-year-old man comes to the emergency department because of a 2-day history of weakness in the lower extremities. He is recovering from a recent respiratory tract illness. Neurological examination shows absent deep tendon reflexes and symmetric weakness in both legs. Laboratory studies show elevated CSF protein and decreased nerve conduction velocity. Which of the following parts of the motor unit is most likely lesioned in this patient?

- A. Axon
- B. Extrafusal fibers
- C. Soma
- D. Myelin sheath
- E. End plate

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0184 Guillain-Barre

Question #: 19

A 62-year-old man comes to the physician because of a 2-hour history of traumatic injury to his left arm. Neurological examination shows manifestations consistent with lower motor neuron injury. Which of the following motor phenomena best represents a negative sign of the injury?

- A. Resting tremor
- B. Intention tremor
- C. Hypotonia
- D. Fasciculations
- E. Waxing

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0180 Peripheral nervous disease manifestations

Question #: 20

A 58-year-old man comes to the physician because of an 8-month history of tremor. Neurological examination shows an expressionless face, resting tremor of the right upper extremity with rigidity. Movements of the right arm and hand are slow. The observed motor deficits most likely reflect

degeneration of cells forming which of the following structures?

- A. Globus pallidus externus
- B. Globus pallidus internus
- C. Caudate nucleus
- D. Putamen
- E. Substantia nigra pars compacta
- F. Substantia nigra pars reticulata
- G. Subthalamus

Attachment:



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Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0190 Parkinson direct and indirect

Question #: 21

A 53-year-old woman comes to the physician because of a 3-day history of weakness. Neurological examination shows decreased power and hyperreflexia in the right upper and lower limbs. Which of the following structures is most likely lesioned?

- A. Left internal capsule (anterior limb)
- B. Right internal capsule (genu)
- C. Left crus cerebri
- D. Right crus cerebri
- E. Right posterior paracentral lobule
- F. Left anterior paracentral lobule

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0153 Correlate upper and lower motor neuron signs with lesion sites

Question #: 22

A 34-year-old man comes to the physician because of a 4-day history of clumsiness affecting the right hand and foot. Physical examination shows right-sided dysdiadochokinesia and past-pointing. Which of the following additional manifestations is most likely?

- A. Athetosis
- B. Chorea
- C. Dystonia
- D. Rigidity
- E. Intention tremor

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 23

A 56-year-old man is brought to the physician by his spouse because of a 4-hour history of uncontrolled rotational movements of his left leg. A diagnosis of hemiballismus is made. Which of the following sites is most likely injured?

- A. Left frontal lobe
- B. Left striatum
- C. Left subthalamus
- D. Left cerebellum
- E. Left tectum
- F. Right frontal lobe
- G. Right striatum
- H. Right subthalamus
- I. Right cerebellum
- J. Right tectum

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0191 Hyperkinesia direct and indirect

Question #: 24

A 48-year-old man comes to the physician because of a 6-hour history of clumsiness. Neurologic examination shows dysarthria, right-sided dysdiadochokinesia, intention tremor, dysmetria and ataxia while attempting to perform skilled movements. Strength and sensory systems are normal. Which of the following cerebellar structures is most likely injured?

- A. Left vestibulocerebellum
- B. Left spinocerebellum
- C. Left cerebrocerebellum
- D. Right vestibulocerebellum
- E. Right spinocerebellum
- F. Right cerebrocerebellum

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 25

A 67-year-old man is brought to the physician because of a 16-year history of Parkinson disease. Which of the following manifestations is/are most likely in this patient?

- A. Dance-like movements
- B. Bradykinesia
- C. Titubation
- D. Athetosis
- E. Intention tremor

Item Description: SOM.MK.II.BPM2.4.NB.2.NSCI.0190 Parkinson direct and indirect

Question #: 26

A 19-year-old woman comes to the physician because of a 3-week history of intermittent blurry vision. Neurologic examination shows an inability of her left eye to adduct during lateral gaze to the right. Both eyes converge normally. MRI of the brain shows an inflammatory lesion affecting the brainstem. Which of the following structures is most likely lesioned in this patient?

- A. Cranial nerve III
- B. Cranial nerve IV
- C. Cranial nerve VI
- D. Medial longitudinal fasciculus
- E. Paramedian pontine reticular formation
- F. Pretectal area

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0235 MLF lesion

Question #: 27

A 6-year-old boy is brought to the physician because of headaches and visual problems for 1 month. Imaging study shows an optic glioma compressing the right optic nerve. Which of the following is the most likely visual deficit caused by this tumor?

- A. Hemianopia
- B. Homonymous anopia
- C. Heteronymous anopia
- D. Quadrantanopia
- E. Monocular blindness

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0228 Visual Pathway Lesion

Question #: 28

A 40-year-old woman comes to the physician because of loss of smell that has lasted for 6 days. She has a history of Covid-19 infection and tested negative 4 days ago. Her vital signs are all within normal limits. Physical examination shows no abnormalities. Which of the following areas is most likely dysfunctional?

- A. Broca
- B. Piriform
- C. Premotor
- D. Primary motor
- E. Somatosensory
- F. Wernicke

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0266 Cortical olfactory areas

Question #: 29

A 39-year-old man comes to the physician because of a 2-day history of blurry vision. Physical examination shows normal appearance of the right eye. Ptosis is present on the left. Neurological examination shows visual acuity of 20/20 in the right eye and 20/200 in the left eye (near card without glasses). Assessment of eye movements shows injury to cranial nerve III. Imaging studies show a subarachnoid aneurysm that is compressing the cranial nerve as it exits the brainstem. Which of the following arteries is most likely expressing the aneurysm?

- A. Anterior cerebral

- B. Middle cerebral
- C. Posterior communicating
- D. Anterior communicating
- E. Vertebral

Item Description: SOM.CS.I.BPM2.1.NB.4.NSCI.0234 Eye movement examination

Question #: 30

A newborn infant is assessed after birth for ambiguous, male-like genitalia. The infant becomes increasingly irritable over the next 4 hours. The blood pressure is remarkably low. Laboratory studies show hypoglycemia, hyponatremia and hyperkalemia. Karyotype studies show 46, XX. Which of the following enzymes is most likely deficient in this patient?

- A. 17-hydroxylase
- B. Aromatase
- C. 21-hydroxylase
- D. 11-beta hydroxylase
- E. Renal 1-alpha hydroxylase

Item Description: Cong adrenal hyperplasia - CYP 21